NADI: Wi-Fi that tells rescuers who is still inside

Universiti Teknologi Malaysia · Huawei ICT Competition 2026–2027, Innovation Track

Live pilot: 1 floor, 3 rooms, by 3 Nov

The problem

In a fire, wardens count people on paper outside. Rescuers go in not knowing who is left.

Every day, aircon keeps cooling rooms with nobody inside.

49%

of a Malaysian office building's energy goes to aircon and fans (ERIA)

Our idea: NADI

Wi-Fi already fills every room. People bend the signal when they sit or move. NADI reads those changes: no cameras, no names.

Level 4, live

Lab 2	Hall A	Room 3
12	0	3
people inside	aircon still ON	people inside

Floor total: 15 people · updated every minute

What it does

Emergency mode
Live headcount per room and floor for wardens and Bomba.

Everyday mode
Flags empty, cooled rooms. Est. waste: RM1,790 per hall a year.

Counts only: no cameras, no names

How we build it AI works in steps 3 and 5 (orange tag). Huawei names in teal. Coded and deployed with Huawei CodeArts.

1. Sense the room
Huawei AP + ESP32
Wi-Fi notices people moving

2. Send the data
Huawei Cloud IoTDA
collects it every minute

3. AI counts people — AI
ModelArts + MindSpore
learns from real headcounts

4. Show it live
ECS, RDS, DCS Redis
live dashboard and history

5. Talk and alert — AI
MaaS AI chat + SMN
ask in BM, alerts to wardens

On the worst day, NADI tells rescuers who is still inside. Every other day, it pays for itself.